DAO CAO THAI

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PROFESSIONAL SUMMARY

Tech Lead & Backend/Data Engineer with 8+ years scaling high-load systems (AaaS, FinTech, Web3) handling millions of requests. Proven track record leading engineering teams up to 10, architecting low-latency data pipelines, and enforcing strict data integrity. Strong foundation in applied math, quant finance, and systems programming (Python, Go, Java).

TECHNICAL SKILLS

Languages: Python, Go, Java, TypeScript, SQL, Lua

Backend & Data: FastAPI, Spring Boot, gRPC, PostgreSQL, Valkey/Redis, ClickHouse, DuckDB, Apache Airflow, Kafka/Event Hubs, PySpark

Cloud & DevOps: Azure AKS, Cloudflare Workers (Durable Objects), Docker, Kubernetes, FluxCD, OpenTofu, MinIO

PROFESSIONAL EXPERIENCE

Venera AI — *Tech Lead / Backend & Data Engineer* *2024 – 2026*

- **Venera API Hub (Tech Lead, team of 10):** Architected an enterprise AI/healthcare API platform integrated with NanoSoft HIS.
- Separated Control Plane and Data Plane using **gRPC** with single-flight pattern; implemented **Valkey + Lua scripts** for sub-millisecond real-time auth, entitlements, and credit validation on the hot path.
- Built asynchronous billing pipelines streaming usage metrics to **Azure Event Hubs** and **ClickHouse** via MinIO Data Lake (Parquet/Avro).
- **Internal ETL Platform (Tech Lead, team of 2):** Designed an Airflow-orchestrated no-code platform utilizing DuckDB and PySpark, cutting dataset onboarding time for medical LLMs by 70%.
- **FHIR Interoperability (Tech Lead, team of 4):** Developed Java/Spring Boot services transforming national BHYT XML data into FHIR R4 resources via FML StructureMaps for hospital integration.

Weegroup — *Backend Developer* *2025 – 2026*

- Engineered a serverless backend on **Cloudflare Edge** (Workers, Durable Objects, KV, D1) for Telegram Mini Apps, serving 1M global active users with zero region-bound server latency.
- Utilized single-writer **Durable Objects** for atomic game-state consistency without database locks, and integrated **TON Connect** for seamless Web3 payment flows.
- Implemented an asynchronous withdrawal pipeline using **RabbitMQ** to decouple blockchain finality from real-time gameplay, handling thousands of daily transactions safely.

6P Capital Management — *Quantitative Developer & Data Engineer* *2021 – 2025*

- Built an end-to-end quant research platform for Vietnamese equities, ingesting real-time tick data via **SSI/VPS WebSockets** into optimized SQL Server data marts.
- Developed a WorldQuant BRAIN-style DSL backed by GPU-accelerated **NumPy/CuPy** rolling-window operators, empowering researchers to run multi-period backtests 5x faster.
- Integrated the **Mosek convex optimization solver** into an automated alpha combination workflow, scaling multi-alpha portfolio construction for institutional deployment.

VMO Holdings — *Backend & Data Engineer* *2019 – 2021*

- Built a metadata-driven self-service Data Integration Platform for Setel (Malaysia fintech) wrapping PipelineWise and Airbyte, automating ingestion pipelines via JSON-driven Airflow DAG generation.

KEY OPEN SOURCE & EDUCATION

- **hitl (Spec-Driven Development CLI):** Created a Claude Code orchestration CLI with 17 custom skills and a Node.js AST indexing engine to lock API contracts and prevent codebase drift.
- **Education:** Foreign Trade University, Bachelor of International Business (2015 – 2019). Focused on Applied AI/Robotics (YOLO, OpenCV, ROS, A* path planning).